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SOME PHYSICAL CONSTANTS*

Speed of light	c	2.998×10^{8} m/s
Gravitational constant	G	6.673×10^{-11} N · m^2/kg^2
Avogadro constant	N_A	6.022×10^{23} mol^{-1}
Universal gas constant	R	8.314 J/mol · K
Mass–energy relation	c^2	8.988×10^{16} J/kg
		931.49 MeV/u
Permittivity constant	ε_0	8.854×10^{-12} F/m
Permeability constant	μ_0	1.257×10^{-6} H/m
Planck constant	h	6.626×10^{-34} J · s
		4.136×10^{-15} eV · s
Boltzmann constant	k	1.381×10^{-23} J/K
		8.617×10^{-5} eV/K
Elementary charge	e	1.602×10^{-19} C
Electron mass	m_e	9.109×10^{-31} kg
Proton mass	m_p	1.673×10^{-27} kg
Neutron mass	m_n	1.675×10^{-27} kg
Deuteron mass	m_d	3.344×10^{-27} kg
Bohr radius	a	5.292×10^{-11} m
Bohr magneton	μ_B	9.274×10^{-24} J/T
		5.788×10^{-5} eV/T
Rydberg constant	R	$1.097\,373 \times 10^{7}$ m^{-1}

*For a more complete list, showing also the best experimental values, see Appendix B.

THE GREEK ALPHABET

Alpha	A	α	Iota	I	ι	Rho	P	ρ
Beta	B	β	Kappa	K	κ	Sigma	Σ	σ
Gamma	Γ	γ	Lambda	Λ	λ	Tau	T	τ
Delta	Δ	δ	Mu	M	μ	Upsilon	Υ	υ
Epsilon	E	ϵ	Nu	N	ν	Phi	Φ	ϕ, φ
Zeta	Z	ζ	Xi	Ξ	ξ	Chi	X	χ
Eta	H	η	Omicron	O	o	Psi	Ψ	ψ
Theta	Θ	θ	Pi	Π	π	Omega	Ω	ω

SOME CONVERSION FACTORS*

Mass and Density

$1\ \text{kg} = 1000\ \text{g} = 6.02 \times 10^{26}\ \text{u}$

$1\ \text{slug} = 14.59\ \text{kg}$

$1\ \text{u} = 1.661 \times 10^{-27}\ \text{kg}$

$1\ \text{kg/m}^3 = 10^{-3}\ \text{g/cm}^3$

Length and Volume

$1\ \text{m} = 100\ \text{cm} = 39.4\ \text{in.} = 3.28\ \text{ft}$

$1\ \text{mi} = 1.61\ \text{km} = 5280\ \text{ft}$

$1\ \text{in.} = 2.54\ \text{cm}$

$1\ \text{nm} = 10^{-9}\ \text{m} = 10\ \text{Å}$

$1\ \text{pm} = 10^{-12}\ \text{m} = 1000\ \text{fm}$

$1\ \text{light-year} = 9.461 \times 10^{15}\ \text{m}$

$1\ \text{m}^3 = 1000\ \text{L} = 35.3\ \text{ft}^3 = 264\ \text{gal}$

Time

$1\ \text{d} = 86\,400\ \text{s}$

$1\ \text{y} = 365\frac{1}{4}\ \text{d} = 3.16 \times 10^7\ \text{s}$

Angular Measure

$1\ \text{rad} = 57.3^\circ = 0.159\ \text{rev}$

$\pi\ \text{rad} = 180^\circ = \frac{1}{2}\ \text{rev}$

Speed

$1\ \text{m/s} = 3.28\ \text{ft/s} = 2.24\ \text{mi/h}$

$1\ \text{km/h} = 0.621\ \text{mi/h} = 0.278\ \text{m/s}$

Force and Pressure

$1\ \text{N} = 10^5\ \text{dyne} = 0.225\ \text{lb}$

$1\ \text{lb} = 4.45\ \text{N}$

$1\ \text{ton} = 2000\ \text{lb}$

$1\ \text{Pa} = 1\ \text{N/m}^2 = 10\ \text{dyne/cm}^2$
$= 1.45 \times 10^{-4}\ \text{lb/in.}^2$

$1\ \text{atm} = 1.01 \times 10^5\ \text{Pa} = 14.7\ \text{lb/in.}^2$
$= 76.0\ \text{cm Hg}$

Energy and Power

$1\ \text{J} = 10^7\ \text{erg} = 0.2389\ \text{cal} = 0.738\ \text{ft} \cdot \text{lb}$

$1\ \text{kW} \cdot \text{h} = 3.6 \times 10^6\ \text{J}$

$1\ \text{cal} = 4.1868\ \text{J}$

$1\ \text{eV} = 1.602 \times 10^{-19}\ \text{J}$

$1\ \text{horsepower} = 746\ \text{W} = 550\ \text{ft} \cdot \text{lb/s}$

Magnetism

$1\ \text{T} = 1\ \text{Wb/m}^2 = 10^4\ \text{gauss}$

*See Appendix D for a more complete list.